Project 2 Design Plan: The Tuxedo Man's Coffin

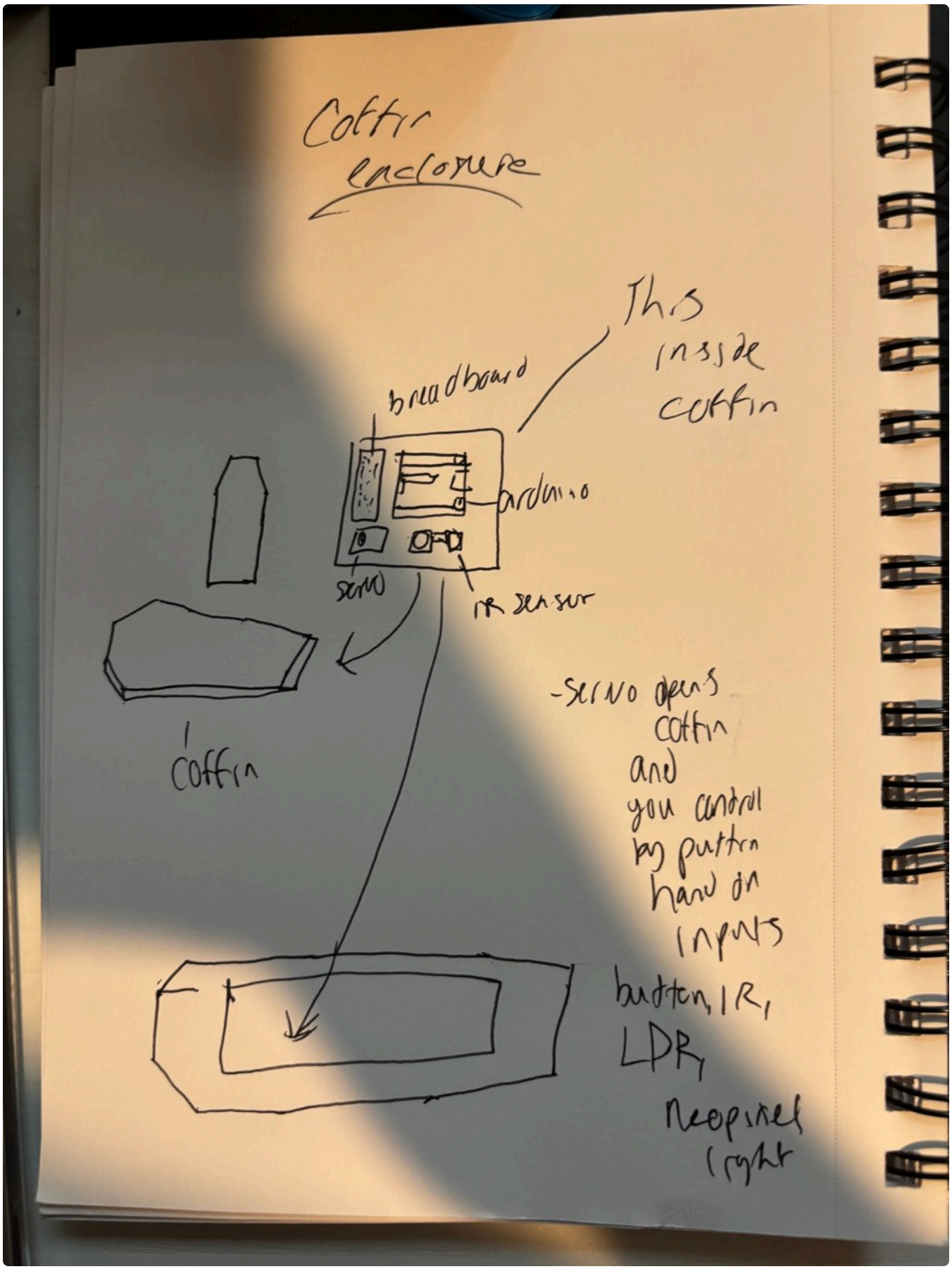
Oct 16, 2026 · @angelo

We're building a laser-cut coffin that knocks from inside when you come close, and opens when you lay your hand on it.

> This is all based on time constraints and not a final idea.

1. Concept

The Tuxedo Man's Coffin is a tabletop coffin (about 36 × 16 × 10.5 cm) that sits on a desk and responds to whoever comes near. It is a physical prop for our Project 1 web game, and everything electronic (Uno, breadboard, servo, sensors) is hidden inside it on a plate on its floor. When someone comes within about 80 cm, something inside starts **knocking**: the servo raps the inside of the head wall and the red light through the lid gap flashes with each knock. A palm print engraved on the side plaque invites you to touch it. Cover it with your hand and the knocking stops, then the lid **creaks open** on a servo with an ember glow inside. Lift your hand and it slams shut. Press the coffin nail on the plaque while the lid is open and it bursts fully open, strobing. It responds to **someone approaching** (the advanced IR sensor) and to **hands on its inputs** (the button and the light sensor, the two simple sensors).


Coffin enclosure
This inside coffin
breadboard
arduino
servo
IR sensor
coffin
-servo opens coffin and you control by putting hand on inputs
button, IR, LDR, Neopixel (right

Concept sketch: the electronics sit inside the coffin. The servo opens the lid and knocks, and you control it by putting your hand on the inputs (button, IR, LDR), with NeoPixel light.

2. Parts plan

Every electronic part is already built and working on our test rig. The only new material is 3 mm plywood from the laser cutter.

Role	Part	Pin / interface	Job in the coffin	Source
Simple sensor 1	Tactile pushbutton, under a laser-cut coffin-nail cap	D2, `INPUT_PULLUP`	Press the nail to make the open coffin burst open	Have it
Simple sensor 2	Photoresistor (LDR) + 10 kΩ voltage divider	A0	Sits in the palm print; a covering hand reads as dark	Have it
Advanced sensor	Sharp GP2Y0D02YK0F IR proximity sensor (~80 cm, digital). **Field guide entry:** "Sharp GP2Y0D02YK0F IR proximity sensor (digital, ~80 cm)" by **Angelo Gonzalez**	A1, analog read against a baseline set at boot (per the entry's gotcha: our unit's HIGH is only ~0.3 V)	Someone is near: starts the knocking	Have it
Output: motion	9 g micro servo ("9g A0090")	D9, `Servo` library	Raps the head wall to knock; pulls a thread to lift the lid	Have it
Output: light	Single WS2812 NeoPixel on its laser-cut round mount	D6	Glows red through the lid gap; strobes on the scare	Have it

Role	Part	Pin / interface	Job in the coffin	Source
Controller	Elegoo UNO R3	USB (power + serial)	Runs the state machine	Have it
Structure	3 mm plywood	Laser cutter	Every structural part (see section 4)	**Laser cutter**

Still to get:

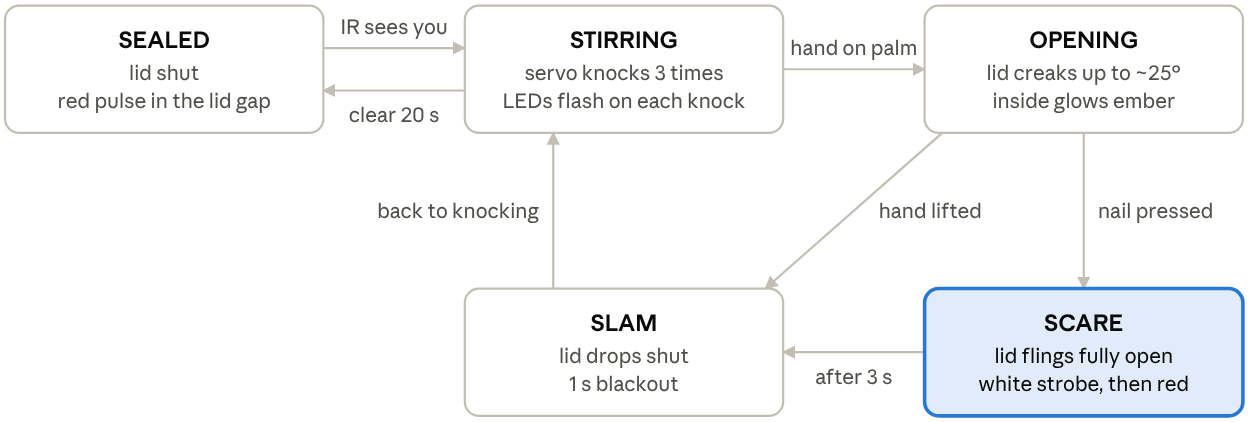
- [ ] 3 mm plywood from the laser cutter: coffin walls, floor, lid, plaque and the electronics plate

3. Behavior spec

The coffin runs six states. The IR sensor decides whether anyone is near; the LDR and the nail button are the visitor's two ways to act on it.

It knocks when you come close and opens only under your hand

Pressing the nail before the lid is open gives one loud knock and nothing else.



state diagram · 6 states

The top row is the approach: noticed, knocking, opening. The bottom row is how it closes again, either when you lift your hand or after the scare.

#	State	Sensor condition	System response
1	**SEALED**	IR clear for 20 s or more	Lid shut. LEDs make a slow red heartbeat pulse that leaks out of the lid gap.
2	**STIRRING**	IR sees someone (3 matching samples, ~120 ms)	Servo arm raps the inside of the head wall 3 times, repeating every 4–6 s at random. The NeoPixel flashes white-red on each knock.
3	**OPENING**	LDR reads dark (hand on the palm print) for 1 s while STIRRING	Knocking stops. The servo winds in its thread and the lid creaks up to ~25° over 3 s; the NeoPixel glows ember orange and flickers.
4	**SLAM**	LDR reads light again while OPENING (hand lifted)	Servo lets the thread go and the lid drops shut, 1 s blackout, then back to STIRRING (or SEALED if IR is clear).
5	**SCARE**	Nail button pressed while OPENING	Servo yanks the lid to full open, the NeoPixel strobes white for 1 s then goes deep red, hold 3 s, then SLAM.
6	Nail too early	Nail pressed while SEALED or STIRRING	One loud knock, a single red flash. No state change.

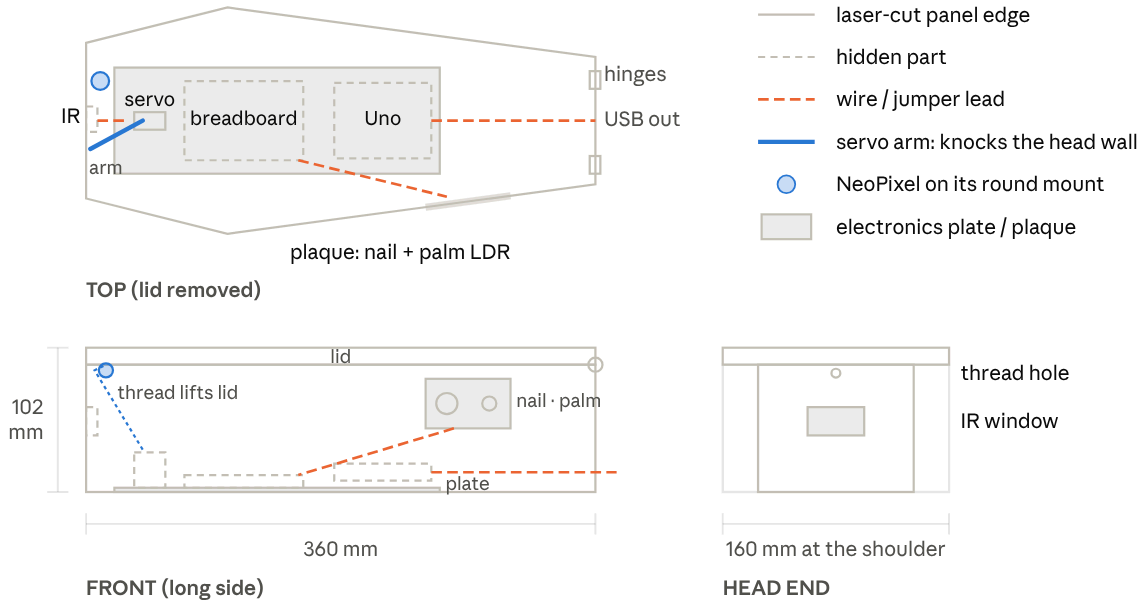
Two calibration rules keep it fair. The LDR's "dark" threshold is set at boot from the room's light, like the IR baseline. The button is debounced (50 ms), so one press is one event.

4. Structure plan

Every structural part is laser-cut from 3 mm plywood, so the structure is effectively 100% digitally fabricated (glue is the only hand work). All the electronics sit on one **electronics plate** on the coffin floor. It's built from the class's common cut patterns: the LKY61 micro-servo cutout, the Arduino Uno mount holes and the 6-post breadboard fence. The lid is hinged at the foot end, so the servo only has to lift the light head end.

All the electronics sit on one plate on the coffin floor

Third-angle orthographic views, 1 : 1.25. Everything laser-cut from 3 mm plywood; lid hinged at the foot end.



orthographic views · top, front, right side

Part	Qty	Notes
Side walls (head, foot, 2 upper, 2 lower)	6	90 mm tall; head wall has the IR window cut-out and a thread hole near the top
Corner gussets	6	Glued inside the angled corners to hold them square
Floor	1	Coffin outline, 360 × 160 mm at the shoulder
Lid	1	Raster-engraved; hinged at the foot end with laser-cut knuckles
Electronics plate	1	230 × 75 mm, class patterns: micro-servo cutout (servo shaft up), Uno holes, breadboard fence slots
Breadboard fence posts	6	Class pattern: 25 × 6 mm with a 15 × 3 mm tab
Uno spacer rings	8	Stacked two per mounting hole so the Uno's solder pins clear the wood

Part	Qty	Notes
Side plaque	1	Engraved palm print with the LDR hole in its centre, plus the nail-button hole
Coffin-nail cap	1	Small disc stack glued onto the button
NeoPixel mount	1	The round mount we already cut; glued under the lid edge at the head end

Wire routing and mechanism: everything runs to the plate on the floor. The button and LDR come off the breadboard on jumper leads and poke out through the side plaque. The IR sensor sits behind the head wall's window, with its lead running along the floor. The NeoPixel's three wires run down the head wall. The servo lies in the plate with its shaft up, so its arm swings sideways: it raps the head wall to knock, and a thread from the arm tip runs up through the head wall to the lid, so winding it in lifts the lid. USB leaves through one hole in the foot panel, away from the visitor.

5. Team plan

We're a team of two. Each area has one owner who makes the final call and keeps it on schedule, but both of us work in every area.

Area	Owner	Backup	Owns
Electronics	@angelo	Alek	Circuit on the plate, jumper leads to the plaque, final install
Code	@angelo	Alek	State machine, IR and LDR calibration, NeoPixel animations, servo knock and lid angles
Fabrication	Alek	Angelo	Laser files (coffin, electronics plate, plaque), cutting, assembly and finish
Documentation	Alek	Angelo	Process photos each session, wiring diagram, final write-up and demo video

Week 9 milestones

Date	Milestone	Done when	Lead
Oct 19, 2026	Laser files final	Coffin, electronics plate and plaque drawn; plate printed at 100% and test-fit with the real servo, Uno and breadboard	Fabrication
Oct 20, 2026	Circuit on the plate	IR, button, LDR, servo and NeoPixel all run together on USB without the board resetting	Electronics
Oct 21, 2026	Coffin cut	Walls, floor, lid and plate cut; dry-fit together	Fabrication
Oct 22, 2026	Behaviors working	Every row of the behavior table works; knock and lid-thread angles set	Code
Oct 23, 2026	Installed and documented	Electronics inside, lid opens on the thread, demo video and photos taken	All

Open questions: we assumed Week 9 is Oct 19–23. Our servo is a 9 g clone, not the class's LKY61, so it gets test-fit on the paper print before we cut.